Assignment 1

Name:

1 Required Exercises

Exercise 1

When compiling the given statement, **7** tokens will be generated. The 1st token is the keyword **int**, the 2nd token is the identifier **a3**, the 3rd token is the operator =, the 4th token is the identifier **a**, the 5th token is the operator *, the 6th token is the integer literal **3**, and the 7th token is the delimiter.

Exercise 2

In a string of length n, there are:

1. $n+1$ prefixes

2. $n-1$ proper prefixes

3. 1 prefix of length m

4. 1 suffix of length m

5. 1 (if $m \neq n$) or 0 (if $m = n$) proper prefix of length m

6. $\frac{n(n+1)}{2}+1$ substrings

7. 2^n subsequences

Exercise 3

1. The language of all strings over the alphabet $\{a, b\}$.

2. The language of all strings over the alphabet $\{a, b\}$ that **end with** $\{aaa, aab, aba, abb\}$.

3. The language of all strings over the alphabet $\{a, b\}$ where character ***b*** appears exactly **3 times**.

Exercise 4

1. $86-0755-[1-9][0-9]^7$

2. $a(a|b)^*b$

3. Let *consonant* → $(b|c|d|f|g|h|j|k|l|m|n|p|q|r|s|t|v|w|x|y|z)^*$, the regular expression is

consonant **a** *consonant* **e** *consonant* **i** *consonant* **o** *consonant* **u** *consonant*.

2 Optional Exercises

Exercise 1

The DFA for the language of all strings over Σ without repeated letters is shown in fig. 1

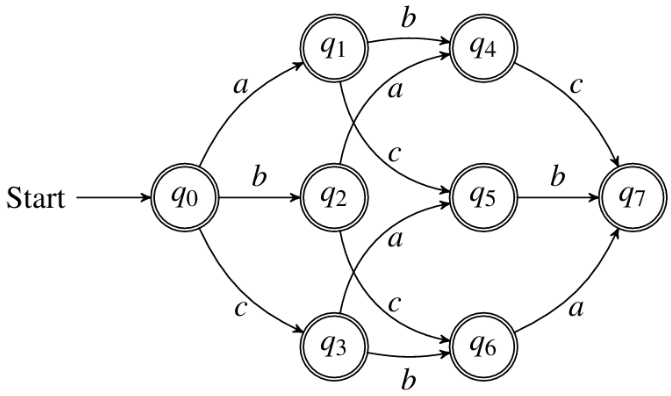

Figure 1: The DFA's transition diagram

According to the transition diagram, the regular expression for the language of all strings over Σ without repeated letters is

$$(\varepsilon|a)(\varepsilon|b)(\varepsilon|c)|$$
$$(\varepsilon|a)(\varepsilon|c)(\varepsilon|b)|$$
$$(\varepsilon|b)(\varepsilon|a)(\varepsilon|c)|$$
$$(\varepsilon|b)(\varepsilon|c)(\varepsilon|a)|$$
$$(\varepsilon|c)(\varepsilon|a)(\varepsilon|b)|$$
$$(\varepsilon|c)(\varepsilon|b)(\varepsilon|a).$$